Theory Assignment :⇒2

1) Assignment Table current schema.

~~Field~~ ~~Attribute~~ Name	Data Type	Key constraint	Null ~~ation~~	Default	Extra
Assignment_ID	Integer	PRI	No	Null	auto_increment
Title	Varchar(255)		NO	Null	
Due_date	Date		NO	Null	

Analysis:

* 1NF (first Normal form): No repeating or multi-valued Attribute
* 2NF (second Normal form): No partial dependency as Assignment_ID is the primary Key.
* 3NF (Third Normal form): No transitive dependency.
* BCNF (Boyce_Codd. Normal form): Fully functional dependencies exist

∴ No need to change!: Table is already BCNF

2) course Table : current schema.

Field	Type	Null	Key	Default	Extra
course_ID	int	No	PRI	Null	
courseName	varchar(255)	No		Null	
instructor_ID	int	Yes	Mul	Null	

Analysis

* BCNF violation:
  - Reason: InstructorID is functionally dependent on course_ID, but an instructor can teach multiple course: This means Instruct_ID deos not solely depend on the primary key

  solution: create a separate table for course_instructor.

Refined Tables::

course: Table.

Field	Type	Null	Key	Default	Extra
course_ID	int	No	PRI	Null	
course_Name	varchar(255)	No		Null	

New course-instructor Table

Field	Type	Null	Key	Default
course_ID	int	No	PRI	Null
Instructor_ID	int	NO	PRI	Null

Now course Table in BCNF

3. Feedback Table (current schema

Field	Type	Null	Key	Default	Extra
feedback_ID	int	No	PRI	Null	
submission_ID	int	YES		Null	
Feedback_Text	Text	YES		Null	

Analysis!
- Already in BCNF:
  - No changes needed!

4) Grade Table (current schema)

Field	Type	Null	Key	Default	Extra
Grade_ID	int	No	PRI	Null	
submission_ID	int	YES		Null	
grade	decimal(5,2)	No		Null	

Analysis
- Grade_ID: is the primary key, and all attributes depends on it directly.

5. Instructor Table: (current schema)

Field	Type	Null	Key	Default	Extra
Instructor_ID	int	No	PRI	NULL	
fname	varchar(100)	No		NULL	
lname	varchar(100)	No		NULL	

Analysis:
- Already in BCNF
- Instructor_ID is the primary key depend on it diretty
  - No changed needed

6. Students Table (current schema)

Field	Type	Null	Key	Default	Extra
student_ID	int	No	PRI	Null	auto_increment
First_Name	Varchar(100)	No		Null	
Email	varchar(100)	No	Null	Null	

Analysis
- stududent_ID is the primary key, and all attributes depend on it directly
  - No change needed.

7. Submission Table (current schema)

	Type	Key	Default	Extra
Submission_ID	int	PRI	NULL	
student_ID	int		NULL	
submission_date	date		NULL	
file_Path	Varchar(255)		NULL	
Assignment_ID	int		NULL	

Analysis:
- Already in BCNF:
- submission_ID is the primary key, and all attributes depend on it directly
- No need to change:

Final Refined schema (All Tables in BCNF)

1) Assignment Table

Field	Type	Null	Key	Default	Extra
Assignment-ID	int	No	PRI	Null	auto_increment
Due-Date	date	No		Null	
Title	varchar(255)	No		Null	

2) course Table:

Field	Type	Null	Key	Default	Extra
course-ID	int	No	PRI	Null	
course-Name	Varchar(255)	No		Null	

3) course ~~Table~~ instructor Table

Field	Type	Null	Key	Default	Extra.
course-ID	int	NO	PRI	Null	
instructor-ID	int	NO	PRI	Null	

4) Feedback Table:

Field					
Feedback-ID	int	NO	PRI	Null	
Submission-ID	int	YES		Null	
Feedback-Text	text	YES		Null	

5) grade Table

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Grade-ID	int	NO	PRI	Null	
Submission-ID	int	YES		Null	
grade	decimal(8,2)	NO		Null	

6) instructor Table.

Instructor-ID	int	NO	PRI	Null	
Fname	varchar(100)	NO		Null	
lname	varchar(100)	NO		Null	

7. Student Table

Field	Type	Null	Key	Default	Extra
student-ID	int	No	PRI	Null	auto_increment
First-Name	varchar(100)	No		Null	
Email	varchar(100)	No	UNI	Null	

8 submission. Table:

Field	Type	Null	Key	Default	Extra
Submission_ID	int	No	PRI	NULL	
Student-ID	int	YES		NULL	
Submission_Date	date	No		NULL	
File Path	varchar(255)	NO		NULL	
Assignment-ID	int	YES		NULL	

Final Thoughts

- All tables are now in BCNF
- minimal redundancy & better data integrity
- More efficient queries & relationships

checking for 4NF (Fourth Normal form)

- 4NF deals with multivalued dependencies (MVDs) if a table has two or more independent multivalued attributes that do not depend on each other, it ~~vala~~ violates 4NF
- None of the tables store multiple independent multivalued attributes
- No table currently violates 4NF.
- All Table are in 4NF

Checking for 5NF (fifth Normal form)

- 5NF deals with join dependencies: if a table can be ~~han~~ decomposed into smaller tables without losing data, it violates 5NF
- our course-instructor table already handles a potential many-to-many relationship
- There are no unnecessary join dependencies

IV All tables are in 5NF